

TASK 2: Sensor-Based Simulation

Project: Automatic LED Control Using LDR (Light Sensor) in Tinkercad

Objective: To turn an LED ON in darkness and OFF in bright light using an LDR sensor.

Components Used:

Arduino Uno, LDR Sensor, LED, 220Ω Resistor, Breadboard, Jumper Wires.

Working Principle:

The LDR detects light intensity. When the light level decreases, the Arduino turns the LED ON. When light intensity increases, the LED turns OFF.

Arduino Code:

```
int ldr=A0;
int led=13;
void setup(){ pinMode(led,OUTPUT); }
void loop(){
  int value=analogRead(ldr);
  if(value<500) digitalWrite(led,HIGH);
  else digitalWrite(led,LOW);
}
```

Simulation Output: LED glows in low light and turns OFF in bright light.